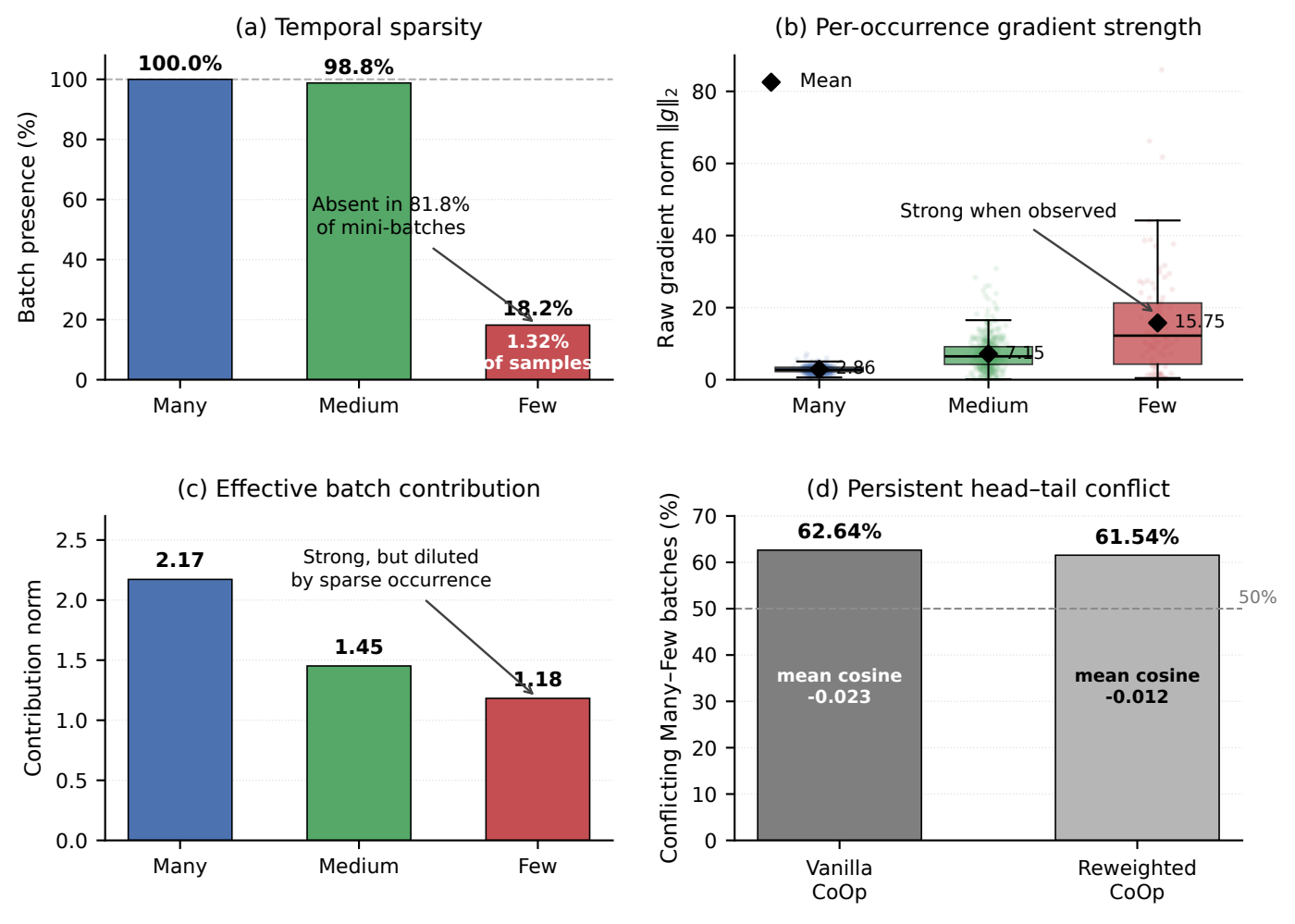
(a) Temporal sparsity
Batch presence (%)
100.0%
98.8%
Absent in 81.8%
of mini-batches
18.2%
1.32%
of samples
Many
Medium
Few
(b) Per-occurrence gradient strength
Raw gradient norm $\|g\|_2$
Mean
Strong when observed
2.86
7.15
15.75
Many
Medium
Few
(c) Effective batch contribution
Contribution norm
2.17
Strong, but diluted
by sparse occurrence
1.45
1.18
Many
Medium
Few
(d) Persistent head–tail conflict
Conflicting Many–Few batches (%)
62.64%
61.54%
50%
mean cosine
-0.023
mean cosine
-0.012
Vanilla
CoOp
Reweighted
CoOp